Optimization of Space Magnetic Gradient Anomaly for Ground-Airborne Frequency-Domain Electromagnetic Detection in Complex Terrain

Shuxu Liu, Jie Liang, and Changsheng Liu

***Abstract*— Ground-airborne frequency-domain electromagnetic (GAFDEM) rapid imaging method holds the potential for swiftly detecting underground anomalies in complex terrains at great depths. However, the current GAFDEM rapid imaging methods have not yet achieved satisfactory detection results in actual complex terrains. This is primarily due to two reasons: 1) the detection environment in complex terrain involves nonhorizontal land and significant relief caused by the presence of mountains. However, current GAFDEM rapid imaging methods fail to consider the influence of terrain on imaging effectiveness, and 2) in controlled-source electromagnetic detection, the design of the observation scheme directly affects the detection effectiveness of imaging methods. In actual detection, terrain constraints limit the design of observation schemes. For GAFDEM, the design of its transmitting sources is greatly influenced by terrain since these sources need to be deployed on the ground. Addressing these issues, this study analyzes the challenges faced by the space magnetic gradient anomaly (SMGA) in complex terrains, which exhibit effective detection under a horizontal ground model. Based on the analysis results, an optimization method for SMGA in complex terrains is proposed. This method achieves rapid correction of the background field in SMGA, thereby enhancing its detection efficacy in complex terrains and enabling the effective identification of underground electrical anomalies. Furthermore, performance testing of the optimized SMGA (O-SMGA) demonstrated that O-SMGA exhibits excellent applicability. Subsequently, an examination is conducted on multiple parameters of the transmitting source assessing their effects on the imaging performance of O-SMGA. Based on the analysis results, design guidelines for GAFDEM transmitting sources in complex terrains applicable to O-SMGA are provided to further enhance its detection effectiveness in actual detection. Finally, a synthetic test of O-SMGA and the transmitting source design guidelines is conducted using models with actual terrain data and actual experimental results. The results indicate that these methods still exhibit excellent performance in actual detection applications.**

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***Index Terms*— Complex terrain, frequency-domain, gradient, ground-airborne frequency-domain electromagnetic (GAFDEM), transmitting source, underground structure detection.**

I. Introduction

DUE to the increasing demand for mountainous area resource development and the construction of mountain tunnels, the need for detecting deep underground structures in complex terrains is also growing. Frequency-domain electromagnetic (FDEM) is an effective technique for rapid detection of underground electrical structures [1], [2], [3]. In FDEM, Ground-Airborne Frequency-Domain Electromagnetic (GAFDEM) uses a ground-based source with airborne data collection, combining the benefits of controlled source audio-frequency magnetotelluric (CSAMT) [4], [5], [6] and airborne frequency-domain electromagnetics (AFEMs) [7], [8], [9]. This method holds the potential for large-depth, large-scale, and efficient underground anomaly detection in complex terrains [10], [11]. The interpretation of GAFDEM data has become a significant research focus. The data interpretation methods of GAFDEM mainly include inversion and rapid imaging. Inversion can effectively reconstruct the distribution of underground structures and physical parameters [12], but it typically requires significant calculation resources and time. Rapid imaging, on the other hand, can also effectively identify underground anomalies, requiring fewer calculation resources and shorter times, although its imaging accuracy is lower than that of inversion. Rapid imaging is beneficial for quick field detection and observation scheme design. Therefore, study on rapid imaging methods for GAFDEM is essential. At present, the relatively mature GAFDEM rapid imaging methods are all proposed based on the horizontal ground model [13], [14], [15]. However, in areas with complex terrain, where the detection environment is nonhorizontal and characterized by significant terrain undulations, the terrain can significantly affect the distribution of GAFDEM's electromagnetic response in the entire space [16]. A study on GAFDEM imaging methods based on a horizontal ground model is likely to be influenced by terrain, thereby potentially reducing the imaging accuracy of underground structures.

In actual detection in complex terrain, besides the influence of terrain on the imaging methods themselves, the design of observation schemes is also subject to the constraints imposed

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by the terrain. Particularly for the transmitting source, terrain factors can restrict the design of parameters such as transmitting source length, transmitting frequency, and transmitting source position. For controlled-source FDEM, designing a suitable transmitting source is crucial [17], [18], [19], [20]. Based on the performance of existing instruments [17], [21], [22], designing various parameters of the transmitting source can theoretically result in better detection results. For GAFDEM, there are only design guidelines for a few parameters of the transmitting source. In the imaging process, apparent resistivity imaging method (ARI) requires the calculation of the skin depth corresponding to different frequency data and uses it as the underground depth corresponding to the imaging data. The calculation of the skin depth requires the transmit-receive distance to satisfy the far-zone. Therefore, in the transmitting source design of ARI, the distance between the transmitting source and the survey area needs to be greater than five times the skin depth [11]. Divergence of tipper vector (DT) borrows the calculation method of the tipper used in the magnetotelluric (MT). The MT method uses natural field sources, which can be equivalent to orthogonal transmitting sources. Therefore, the transmitting source for DT is required to be an orthogonal source [13], [14]. Meanwhile, in complex terrains, the terrain conditions are more complex. Certain terrains may restrict the design of certain transmitting source parameters. For example, parameters such as transmitting source length and direction cannot be arbitrarily set.

Therefore, to enhance the underground anomaly detection capability of GAFDEM in areas with complex terrains, this study proposes an optimization method for space magnetic gradient anomaly (SMGA) to such terrain conditions, based on its proven effectiveness in underground anomaly detection on flat terrain. The optimization method comprises two main components: optimization of the SMGA algorithm and optimization of transmitting source design in the survey scheme. The optimization of the SMGA algorithm first identifies the issue of background field calculation errors in complex terrain, to address this issue, a method is developed to calculate the background field. This method corrected the background field calculation error and significantly improved calculation speed. Using this optimized background field, the SMGA imaging results can be corrected to improve accuracy and enhance imaging efficiency. The optimization of transmitting source design analyzes the impact of various parameters of the transmitting source on the imaging performance of optimized SMGA (O-SMGA) for a fixed model and, based on the analysis results, summarizes the guidelines for optimizing transmitting source design for SMGA in complex terrain. By optimizing the transmitting source according to this guideline, the O-SMGA's ability to detect anomalies in complex terrain can be further enhanced. This study provides new techniques for real-time GAFDEM detection in complex terrains.

II. Optimization of SMGA Algorithm in Complex Terrain

A. Principle of SMGA

The survey data of GAFDEM is the magnetic field along airborne survey lines. This magnetic field is generated by the transmitting source exciting the underground medium and decays with increasing distance from the transmitting source, with the decay rate depending on the physical parameters of the underground medium [23]. When the earth is of purely electrical structure, the decay rate of this magnetic field can reflect the distribution of underground electrical structures in the surveyed area. Gradient, on the other hand, is a parameter reflecting the rate of change of physical quantities. Therefore, the gradient of this magnetic field can characterize underground electrical structures. However, when using the gradient of GAFDEM to identify electrical anomaly structures in the ground, the electrical properties of the background earth can cause the gradient of this magnetic field to have a baseline, making it difficult to effectively identify the position and boundaries of underground electrical anomalies. Therefore, SMGA is proposed for identifying underground electrical anomaly structures. SMGA is a 3-D method of interpreting GAFDEM data that utilize the difference between the actual magnetic field gradient and the background magnetic field gradient to identify changes in electrical properties of underground media. This method corrects the influence of the gradient of the magnetic field caused by the electrical properties of the background earth, directly reflecting the degree of difference between the underground electrical structure and the background earth's electrical structure. The calculation method of SMGA is shown in Section II-A.

First, it is necessary to calculate the space gradient of the actual magnetic field. A method for calculating the space gradient of the actual measured magnetic field, denoted as *Grad*, is defined as shown in (1), at the bottom of the next page, with a dimension of $T/m/\log_{10}(\text{Hz})$. $B(x_i, y_i, f_k)$ is the z-component of the actual magnetic field at the survey point with horizontal positions (x_i, y_i) at transmitting frequency f_k. The variable $i = 1, 2, \ldots, p$ denotes the survey point index, while $k = 1, 2, \ldots, q$ denotes the transmitting frequency index. Equation (1) normalizes the difference in magnetic field between two points at different frequencies based on the horizontal distance and the difference of the frequency logarithm between the two points to obtain the space gradient of the actual magnetic field *Grad*. *Grad* characterizes the rate of change of the actual magnetic field with respect to both survey points and frequencies. Furthermore, by substituting $B(x_i, y_i, f_k)$ in (1) with the background magnetic field z-component $B_0(x_i, y_i, f_k)$, the equation for calculating the space gradient of the background magnetic field, denoted as $Grad_0$, can be obtained as shown in (2), at the bottom of the next page, with a dimension of $T/m/\log_{10}(\text{Hz})$. $Grad_0$ characterizes the rate of change of the background magnetic field with respect to both survey points and frequencies. When the background resistivity of the ground is known, the GAFDEM 1-D forward modeling can be used to calculate $B_0(x_i, y_i, f_k)$ [11]. The fitting method for background resistivity is based on study into horizontal ground SMGA [15]. Finally, it is necessary to calculate the absolute deviation $\Delta Grad$ between the gradient of the actual magnetic field and the gradient of the background magnetic field, as shown in (3), at the bottom of the next page, with a dimension of $T/m/\log_{10}(\text{Hz})$. $\Delta Grad$ characterizes the variations in underground electrical structure.

A single-frequency $\Delta Grad$ can be used to identify 2-D underground electrical structures in the horizontal direction.

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However, in GAFDEM applications, identifying anomalies in the vertical direction, that is, recognizing the depth of the anomaly, is crucial. GAFDEM typically identifies the depth of an anomaly by analyzing the variation in imaging parameters across different frequencies. However, $\Delta Grad$ alone cannot achieve this depth recognition due to its frequency-dependent characteristic. In GAFDEM, the magnetic field generated by a transmitting source with the same electric dipole moment at a given point increases with frequency. As a result, the space magnetic field gradient at all points on a survey line will increase with rising frequencies. Consequently, the absolute deviation between the actual magnetic field gradient and the background magnetic field gradient $\Delta Grad$ also tends to increase with frequency, thereby limiting its ability to distinguish the depth of anomalies. To correct for this frequency-related effect on $\Delta Grad$, first, the frequency gradient of the background magnetic field, $grad_0$, is defined to capture the frequency-dependent characteristics of the gradient, as shown in (4) with a dimension of $T/m/\log_{10}(\mathrm{Hz})$. Based on $grad_0$, a correction factor Cor is defined in (5), which is a dimensionless parameter. Cor encapsulates the frequency-dependent characteristics of the background magnetic field gradient, allowing $\Delta Grad$ for normalization across different frequencies to obtain SMGA, as in (6) with a dimension of $T/m/\log_{10}(\mathrm{Hz})$. In theory, any differences between SMGA at varying frequencies are due solely to variations in underground electrical structures, enabling the recognition of depth in electrical anomalies

$$\mathrm{grad}_0(x_i, y_i, f_k) = \left\{ \begin{array}{l} \frac{B_0(x_i,y_i,f_2)-B_0(x_i,y_i,f_1)}{\log_{10} f_2-\log_{10} f_1}(k=1) \\ \frac{B_0(x_i,y_i,f_{k+1})-B_0(x_i,y_i,f_{k-1})}{\log_{10} f_{k+1}-\log_{10} f_{k-1}}(1<k<q) \\ \frac{B_0(x_i,y_i,f_q)-B_0(x_i,y_i,f_{q-1})}{\log_{10} f_q-\log_{10} f_{q-1}}(k=q) \end{array} \right\} \tag{4}$$

$$\mathrm{Cor}(x_i, y_i, f_k) = \frac{\mathrm{grad}_0(x_i, y_i, f_k)}{\sum_{i=1}^{p}\sum_{k=1}^{q}\mathrm{grad}_0(x_i, y_i, f_k)} \tag{5}$$

$$\mathrm{SMGA}(x_i, y_i, f_k) = \mathrm{Cor}(x_i, y_i, f_k)\Delta\mathrm{Grad}(x_i, y_i, f_k) \tag{6}$$

where SMGA is defined based on the horizontal positions and frequencies of survey points. When imaging underground structures, it is necessary to map SMGA to corresponding underground positions. The horizontal coordinates of SMGA are the same as the horizontal coordinates of the survey points. The vertical position, denoted as z, is defined using the skin depth, as shown in (7) with a dimension of m. It represents the maximum depth at which electromagnetic waves at that frequency can propagate in the medium. When the distance between the transmitting source and the survey line satisfies the far-zone condition, the calculation of the skin depth is only related to the electrical resistivity, magnetic permeability of the medium, and the frequency of the electromagnetic waves [24]. In (7), ρ_0 represents the background electrical resistivity, and μ denotes the background magnetic permeability. According to Maxwell's equations, variations in a medium's resistivity, magnetic permeability, and dielectric constant all contribute to changes in the electromagnetic field in space. In actual detection, the resistivity, magnetic permeability, and dielectric constant usually vary from one location to another because of differences in underground media. However, for GAFDEM rapid imaging methods, changes in magnetic permeability and dielectric constant are not the primary focus; instead, variations in resistivity are the main target. In GAFDEM studies, models typically set the magnetic permeability of the entire space to the vacuum permeability and the dielectric constant to the vacuum dielectric constant. Thus, in (7), μ represents the vacuum permeability. When calculating z in complex terrain, due to the relief of the terrain, the altitude directly beneath the

$$\mathrm{Grad}(x_i, y_i, f_k) = \left\{ \begin{array}{l} \frac{[B(x_2,y_2,f_2)-B(x_2,y_2,f_1)]-[B(x_1,y_1,f_2)-B(x_1,y_1,f_1)]}{\sqrt{(y_2-y_1)^2+(x_2-x_1)^2}(\log_{10} f_2-\log_{10} f_1)}(i=1,k=1) \\ \frac{[B(x_2,y_2,f_{k+1})-B(x_2,y_2,f_{k-1})]-[B(x_1,y_1,f_{k+1})-B(x_1,y_1,f_{k-1})]}{\sqrt{(y_2-y_1)^2+(x_2-x_1)^2}(\log_{10} f_{k+1}-\log_{10} f_{k-1})}(i=1,1<k<q) \\ \frac{[B(x_{i+1},y_{i+1},f_2)-B(x_{i+1},y_{i+1},f_1)]-[B(x_{i-1},y_{i-1},f_2)-B(x_{i-1},y_{i-1},f_1)]}{\sqrt{(y_{i+1}-y_{i-1})^2+(x_{i+1}-x_{i-1})^2}(\log_{10} f_2-\log_{10} f_1)}(1<i<p,k=1) \\ \frac{[B(x_{i+1},y_{i+1},f_{k+1})-B(x_{i+1},y_{i+1},f_{k-1})]-[B(x_{i-1},y_{i-1},f_{k+1})-B(x_{i-1},y_{i-1},f_{k-1})]}{\sqrt{(y_{i+1}-y_{i-1})^2+(x_{i+1}-x_{i-1})^2}(\log_{10} f_{k+1}-\log_{10} f_{k-1})}(1<i<p,1<k<q) \\ \frac{[B(x_p,y_p,f_{k+1})-B(x_p,y_p,f_{k-1})]-[B(x_{p-1},y_{p-1},f_{k+1})-B(x_{p-1},y_{p-1},f_{k-1})]}{\sqrt{(y_p-y_{p-1})^2+(x_p-x_{p-1})^2}(\log_{10} f_{k+1}-\log_{10} f_{k-1})}(i=p,1<k<q) \\ \frac{[B(x_{i+1},y_{i+1},f_q)-B(x_{i+1},y_{i+1},f_{q-1})]-[B(x_{i-1},y_{i-1},f_q)-B(x_{i-1},y_{i-1},f_{q-1})]}{\sqrt{(y_{i+1}-y_{i-1})^2+(x_{i+1}-x_{i-1})^2}(\log_{10} f_q-\log_{10} f_{q-1})}(1<i<p,k=q) \\ \frac{[B(x_p,y_p,f_q)-B(x_p,y_p,f_{q-1})]-[B(x_{p-1},y_{p-1},f_q)-B(x_{p-1},y_{p-1},f_{q-1})]}{\sqrt{(y_p-y_{p-1})^2+(x_p-x_{p-1})^2}(\log_{10} f_q-\log_{10} f_{q-1})}(i=p,k=q) \end{array} \right\} \tag{1}$$

$$\mathrm{Grad}_0(x_i, y_i, f_k) = \left\{ \begin{array}{l} \frac{[B_0(x_2,y_2,f_2)-B_0(x_2,y_2,f_1)]-[B_0(x_1,y_1,f_2)-B_0(x_1,y_1,f_1)]}{\sqrt{(y_2-y_1)^2+(x_2-x_1)^2}(\log_{10} f_2-\log_{10} f_1)}(i=1,k=1) \\ \frac{[B_0(x_2,y_2,f_{k+1})-B_0(x_2,y_2,f_{k-1})]-[B_0(x_1,y_1,f_{k+1})-B_0(x_1,y_1,f_{k-1})]}{\sqrt{(y_2-y_1)^2+(x_2-x_1)^2}(\log_{10} f_{k+1}-\log_{10} f_{k-1})}(i=1,1<k<q) \\ \frac{[B_0(x_{i+1},y_{i+1},f_2)-B_0(x_{i+1},y_{i+1},f_1)]-[B_0(x_{i-1},y_{i-1},f_2)-B_0(x_{i-1},y_{i-1},f_1)]}{\sqrt{(y_{i+1}-y_{i-1})^2+(x_{i+1}-x_{i-1})^2}(\log_{10} f_2-\log_{10} f_1)}(1<i<p,k=1) \\ \frac{[B_0(x_{i+1},y_{i+1},f_{k+1})-B_0(x_{i+1},y_{i+1},f_{k-1})]-[B_0(x_{i-1},y_{i-1},f_{k+1})-B_0(x_{i-1},y_{i-1},f_{k-1})]}{\sqrt{(y_{i+1}-y_{i-1})^2+(x_{i+1}-x_{i-1})^2}(\log_{10} f_{k+1}-\log_{10} f_{k-1})}(1<i<p,1<k<q) \\ \frac{[B_0(x_p,y_p,f_{k+1})-B_0(x_p,y_p,f_{k-1})]-[B_0(x_{p-1},y_{p-1},f_{k+1})-B_0(x_{p-1},y_{p-1},f_{k-1})]}{\sqrt{(y_p-y_{p-1})^2+(x_p-x_{p-1})^2}(\log_{10} f_{k+1}-\log_{10} f_{k-1})}(i=p,1<k<q) \\ \frac{[B_0(x_{i+1},y_{i+1},f_q)-B_0(x_{i+1},y_{i+1},f_{q-1})]-[B_0(x_{i-1},y_{i-1},f_q)-B_0(x_{i-1},y_{i-1},f_{q-1})]}{\sqrt{(y_{i+1}-y_{i-1})^2+(x_{i+1}-x_{i-1})^2}(\log_{10} f_q-\log_{10} f_{q-1})}(1<i<p,k=q) \\ \frac{[B_0(x_p,y_p,f_q)-B_0(x_p,y_p,f_{q-1})]-[B_0(x_{p-1},y_{p-1},f_q)-B_0(x_{p-1},y_{p-1},f_{q-1})]}{\sqrt{(y_p-y_{p-1})^2+(x_p-x_{p-1})^2}(\log_{10} f_q-\log_{10} f_{q-1})}(i=p,k=q) \end{array} \right\} \tag{2}$$

$$\Delta\mathrm{Grad}(x_i, y_i, f_k) = \mathrm{Grad}(x_i, y_i, f_k) - \mathrm{Grad}_0(x_i, y_i, f_k) \tag{3}$$

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survey point should be used as the reference. Therefore, terrain information is incorporated into the calculation of z. Here, $Z(i)$ represents the elevation directly beneath each survey point. In actual detection, this information can be obtained using specialized geographic information system (GIS) software such as Bigemap GIS Office (BIGEMAP). Through (7), SMGA can be correlated with the underground positions during imaging, as shown in (8). Furthermore, to better illustrate the distribution of different underground electrical structures, (8) also normalizes SMGA based on its maximum absolute value to produce a dimensionless imaging parameter.

$$z_i = Z_i - \sqrt{\rho_0/(\pi f_k \mu)} \tag{7}$$

$$\mathrm{SMGA}(x_i, y_i, z_i) = \frac{\mathrm{SMGA}(x_i, y_i, f_k)}{\max(|\mathrm{SMGA}|)}. \tag{8}$$

B. Feasibility Analysis of SMGA in Complex Terrain

To analyze the feasibility of SMGA in complex terrains, a simplified mountainous model in Fig. 1 is constructed using COMSOL multiphysics (COMSOL). The parameters of the model, including grid, materials, etc., are detailed in the appendix. In Fig. 1, the red line represents the transmitting source, centered at the origin of coordinates. The transmitting current has an amplitude of 40 A. The transmitting frequency ranges from 32 to 1024 Hz with an interval of 4 Hz. The mountain is pyramid-shaped with a height of 1000 m. The upper and lower surfaces are squares with side lengths of 1000 and 2000 m, respectively. The blue lines denote the cuboid-shaped anomalies, as shown in Fig. 1. Two scenarios of anomalous resistivity are considered: low-resistivity anomaly $\rho_1 = 10$ Ωm and high-resistivity anomaly $\rho_1 = 1000$ Ωm. A straight line along $x = 0$ m, $y = 4000$–6000 m, $z = 1100$ m is chosen as the survey line for this model, as indicated by the green line in the figure. In the study of SMGA, a step length of 1 m was used to calculate the complex magnetic field z-component at each survey point for every transmitting frequency as the data for GAFDEM. However, in actual detection, the flying platforms carrying electromagnetic receiving systems often travel at high speeds and employ the method of long-term signal stacking to suppress random noise in the measured signals [25]. As a result, the spacing between survey points along the survey line is often much larger than 1 m. Additionally, the hollow coil sensors commonly used in GAFDEM inherently exhibit inductive and capacitive loads, leading to phase shifts in the measured signals [26], [27], thus compromising the accuracy of phase measurements of the measured magnetic field. Therefore, to make the feasibility analysis of SMGA more closely resemble actual surveys, a step length of 10 m was adopted to calculate the amplitude of the magnetic field z-component at each survey point for every transmitting frequency using COMSOL as the data for GAFDEM. Fig. 2 illustrates the imaging results of the model in Fig. 1 using SMGA. The dashed lines in Fig. 2 indicate the positions and boundaries of the anomalies.

In the horizontal ground model, SMGA primarily identifies electrical anomalies through abnormal regions. In the imaging process, a threshold is set to delineate the abnormal regions. In the imaging results, an abnormal region is depicted as a

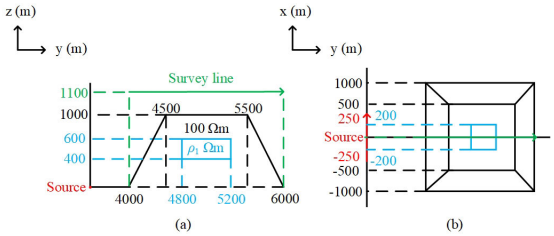


Fig. 1. Model of GAFDEM in simplified mountainous terrain. (a) yz profile and (b) xy profile.

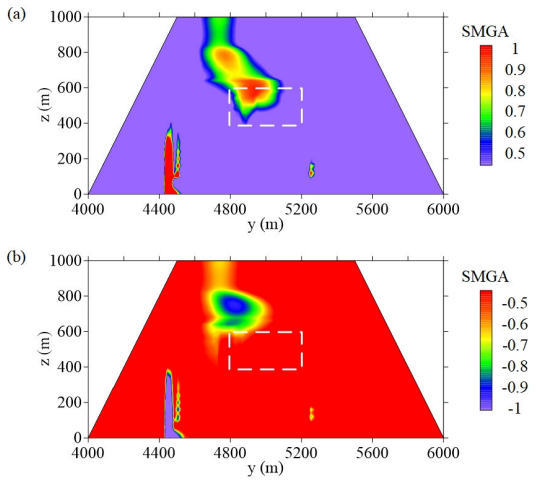


Fig. 2. Imaging results of SMGA for the model in Fig. 1. (a) Low-resistance anomaly and (b) high-resistance anomaly.

closed region with a different color from most other regions, indicating a significant deviation in composite resistivity from the background resistivity. This suggests the presence of electrical anomalies at those locations. Specifically, the positions and boundaries of abnormal regions in the magnitude are used to identify the locations and boundaries of underground electrical anomalies, while the positive or negative values of SMGA in the real and imaginary parts of abnormal regions are utilized to distinguish between different types of electrical anomalies. Since only the magnitude of the magnetic field is used in the imaging process in Fig. 2, the SMGA are real numbers, and thus, there are no longer imaging results for the real and imaginary parts of SMGA. From Fig. 2, it can be observed that abnormal regions exist. The positive and negative values of SMGA in the abnormal regions of the low-resistivity anomaly model and the high-resistivity anomaly model are opposite. However, there are significant disparities in the positions and boundaries of abnormal regions compared to the anomalies in the model shown in Fig. 1. Additionally, false anomalies appear in the deeper layers. It is concluded that this imaging result cannot effectively identify underground anomalies, indicating that the existing SMGA is not entirely suitable for complex terrains with significant topographical variations.

C. Optimization of SMGA in Complex Terrain

According to Zhang et al.'s [16] study, the relief of terrain affects the distribution of the GAFDEM electromagnetic field throughout space. However, the GAFDEM 1-D forward

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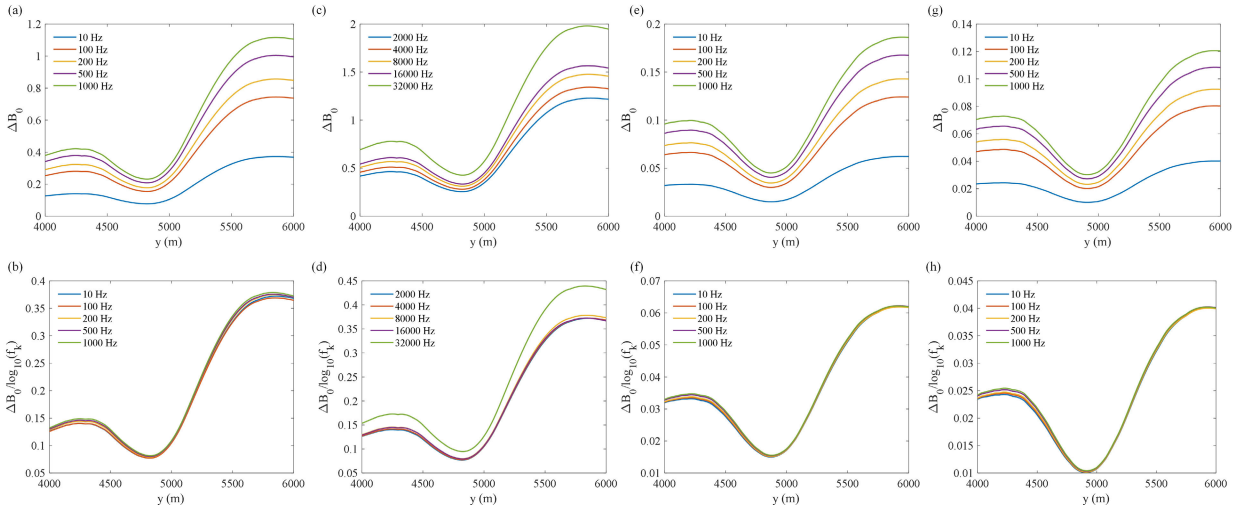


Fig. 3. Comparison between the background magnetic fields calculated using GAFDEM forward modeling and those calculated using COMSOL for the model in Fig. 1. The background resistivity is set to 100 Ωm, with frequencies ranging from 10 to 1000 Hz. (a) Relative deviation ΔB_0. (b) Normalized relative deviation $\Delta B_0/\log_{10}(f_k)$. The background resistivity is set to 100 Ωm, with frequencies ranging from 2000 to 32 000 Hz. (c) Relative deviation ΔB_0. (d) Normalized relative deviation $\Delta B_0/\log_{10}(f_k)$. The background resistivity is set to 1000 Ωm, with frequencies ranging from 10 to 1000 Hz. (e) Relative deviation ΔB_0. (f) Normalized relative deviation $\Delta B_0/\log_{10}(f_k)$. The background resistivity is set to 2000 Ωm, with frequencies ranging from 10 to 1000 Hz. (g) Relative deviation ΔB_0. (h) Normalized relative deviation $\Delta B_0/\log_{10}(f_k)$.

modeling used to calculate the background magnetic field for SMGA can only be applied in horizontal ground models. This limitation leads to differences between the background magnetic field used in the imaging process and the actual background magnetic field, ultimately resulting in deviations between the imaging results and the model settings. The most effective solution to this issue is to utilize COMSOL to construct a model with the same relief in Fig. 1, excluding underground anomalies, and calculate the background magnetic field for each frequency. However, in actual detection, SMGA serves as a rapid imaging method requiring the ability to conduct on-site imaging quickly and guide the design of field survey schemes. Due to the inherent constraints of the finite element method, employing COMSOL to load actual terrain and calculate the background magnetic field for each frequency will demand substantial calculation resources and long times, making it unsuitable for real-time field surveys. Therefore, to further enhance the applicability of SMGA in actual detection and improve the efficiency of background magnetic field calculation, an analysis is conducted on the relationship between the background magnetic field calculated using the GAFDEM forward modeling and that calculated using COMSOL

$$\Delta B_0(x_i, y_i, f_k) = \frac{B_{01}(x_i, y_i, f_k) - B_0(x_i, y_i, f_k)}{B_0(x_i, y_i, f_k)}. \quad (9)$$

A method for calculating the relative deviation of the background magnetic field obtained by the two methods is defined as shown in (9). This relative deviation is dimensionless. $B_0(x_i, y_i, f_k)$ is the background magnetic field calculated using the GAFDEM 1-D forward modeling, and $B_{01}(x_i, y_i, f_k)$ is the background magnetic field calculated using COMSOL. Using the survey line in Fig. 1 as the study subject, the background magnetic fields at transmitting frequencies of 10, 100, 200, 500, and 1000 Hz are separately calculated using both methods. The relative anomalies ΔB_0 are calculated using (9) and plotted for different frequencies, as shown in Fig. 3(a). It can be observed from the figure that the distribution pattern of ΔB_0 at different frequencies along the entire survey line is similar, but ΔB_0 at the same survey point increases with frequency. Based on this characteristic, an attempt is made to normalize ΔB_0 using a frequency-related parameter. $\Delta B_0/\log_{10}(f_k)$ is plotted, as shown in Fig. 3(b). It can be seen from the figure that the curves of $\Delta B_0/\log_{10}(f_k)$ for different frequencies overlap. This suggests that, in actual detection, it is unnecessary to calculate the background magnetic field for every frequency using finite element methods such as COMSOL. Instead, after obtaining the actual terrain information of the survey area using GIS software, a simplified model without anomalies can be established using finite element methods such as COMSOL, and the actual background magnetic field of the survey line at a specific frequency f_0 can be calculated. Furthermore, based on the background magnetic field calculated using COMSOL and GAFDEM 1-D forward modeling, the relative anomaly $\Delta B_0/\log_{10}(f_0)$ for that frequency can be obtained. Finally, the actual background magnetic field B_1 for other frequencies can be rapidly calculated, as shown in the following equation with a dimension of T

$$B_1(x_i, y_i, f_k) = \frac{\Delta B_0(x_i, y_i, f_0) B_0(x_i, y_i, f_k) \log_{10}(f_k)}{\log_{10}(f_0)} + B_0(x_i, y_i, f_k). \quad (10)$$

To analyze the applicability of this background calculation method, the study subject is shifted to the background magnetic fields at transmitting frequencies of 2000, 4000, 8000, 16 000, and 32 000 Hz. The curves of ΔB_0 and $\Delta B_0/\log_{10}(f_k)$ are redrawn and represented in Fig. 3(c) and (d), respectively. From Fig. 3(d), it can be observed that the $\Delta B_0/\log_{10}(f_k)$

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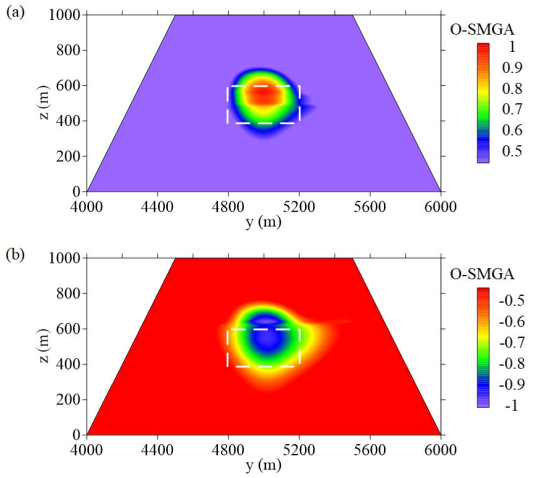


Fig. 4. After correcting the background magnetic field, the imaging results of the O-SMGA for the model in Fig. 1. (a) Low-resistance anomaly. (b) High-resistance anomaly.

at 2000, 4000, 8000, and 16 000 Hz exhibit relatively high normalization accuracy, consistent with those at 10–1000 Hz. The $\Delta B_0/\log_{10}(f_k)$ at 32 000 Hz is comparatively larger. Considering both Fig. 3(b) and (d), it is concluded that this background field calculation method performs well within the frequency band of 10–16 000 Hz but shows reduced accuracy beyond this band. For GAFDEM, the commonly used transmitting frequency band is 16–8192 Hz. Therefore, this background field calculation method meets the requirements of GAFDEM in terms of frequency. Next, the impact of background resistivity variation on this method is analyzed. The background resistivity of the model in Fig. 1 is changed to 1000 and 2000 Ωm, and the curves of ΔB_0 and $\Delta B_0/\log_{10}(f_k)$ are redrawn at 10–1000 Hz, as shown in Fig. 3(e)–(h). It can be seen from Fig. 3(e)–(h) that even after changing the background resistivity, $\Delta B_0/\log_{10}(f_k)$ still exhibits relatively high normalization accuracy. This result indicates that the effect of background resistivity variation on this background field calculation method can be disregarded. Finally, to further validate the effectiveness of the proposed background field calculation method, the model in Fig. 1 is used as an example, and COMSOL is employed to calculate the background magnetic field at $f_0 = 10$ Hz along the survey line, resulting in B_{01}. Using (9) and (10), the background fields at all frequencies B_1 are calculated. By replacing B_0 in the SMGA calculation with B_1 and imaging, the imaging results is shown in Fig. 4.

Comparing Figs. 2 and 4 reveals that after correcting the background magnetic field, the positions of abnormal regions in the imaging results correspond to the anomalies in the model in Fig. 1. The boundaries of the abnormal regions match those of the anomalies. In the low-resistivity anomaly model, the abnormal regions appear positive, while in the high-resistivity anomaly model, they appear negative. Based on this characteristic, the type of electrical anomalies can be identified. In Fig. 4, false anomalies are also eliminated. In comparison with the imaging results of SMGA in the horizontal ground model, the recognition ability of the anomaly boundaries in Fig. 4's imaging results is slightly inferior. This is attributed to the fact that in the SMGA study in the horizontal ground model, complex magnetic fields are used. The phase of the magnetic field contains some information about the underground electrical structures [28], [29]. In the imaging process of Fig. 4, to better match the instrument in actual detection, only the amplitude of the magnetic field is used to calculate SMGA. As a result, the SMGA lacks some of the underground electrical structure information contained in the phase, weakening its ability to identify the boundaries of anomalies. However, Fig. 4's imaging results still accurately identify the central position and type of electrical anomalies. Therefore, within a certain tolerance range, the O-SMGA can be preliminarily considered effective for identifying underground electrical anomalies in complex terrains. Furthermore, taking a computer with an Intel I7-12700KF CPU as an example, the calculation of the background magnetic field of the model in Fig. 1 using the ac/dc module of COMSOL at a frequency of 100 Hz requires a computer memory of 40.3 GB and the time of 5 min and 32 s (t_1). The model's mesh and other settings are shown in the appendix. The number of transmitting frequencies (n) in Fig. 1 model is 249. If the background magnetic field for all frequencies are calculated using COMSOL, without considering changes in mesh with frequency, the time required to calculate the background magnetic field for all frequencies would be at least nt_1. However, with the background field calculation method proposed in this section, using MATLAB to program the algorithm, the calculation of the background magnetic field for one frequency only requires 5 s (t_2). The time required to calculate the background magnetic field for all frequencies using this method is approximately $t_1 + (n-1)t_2$. This method is at least $(n-1)(t_1-t_2)$ faster than the calculation time of COMSOL [30], [31]. Moreover, in actual detection, the complexity of the terrain is much greater than that of the model in Fig. 1, which would further increase t_1. Meanwhile, t_2, being an analytical calculation, remains unaffected. Therefore, it can be concluded that compared to calculating the background magnetic field for all frequencies using COMSOL, the background field calculation method proposed in this section can significantly enhance calculation speed, thereby improving imaging efficiency and aiding in rapid field detection.

Based on the content of this section, the imaging process of O-SMGA is shown in Fig. 5. First, it is necessary to obtain the magnetic field z-component B along the survey line and the terrain information Z_1 of the surveyed area. Following the methodology employed in horizontal ground SMGA studies, the background resistivity fitting method is utilized to calculate the background resistivity ρ_0 based B, and the uncorrected background magnetic field B_0 is calculated using the GAFDEM 1-D forward modeling. By integrating ρ_0 and Z_1, the actual surveyed area is modeled in finite element software such as COMSOL, and the background magnetic field B_{01} at the specified frequency f_0 is calculated. Subsequently, the background magnetic field B_0 is corrected to obtain the corrected field B_1. Finally, O-SMGA is calculated, and the positional information of the survey points and frequency information are converted into imaging positional information.

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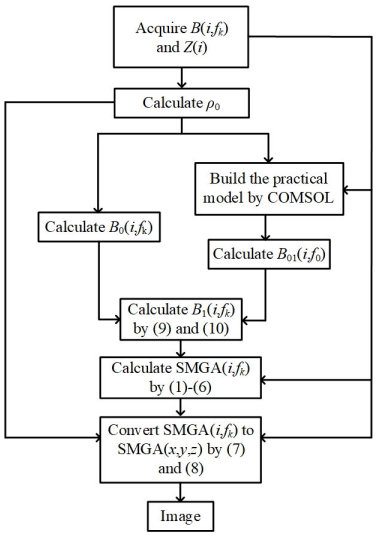


Fig. 5. Imaging flow of O-SMGA.

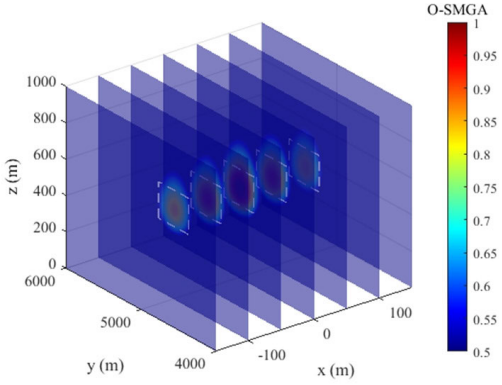


Fig. 6. Three-dimensional detection capability test: imaging results of O-SMGA for the model in Fig. 1 after adding survey lines.

D. Performance Testing of O-SMGA

This section evaluates certain aspects of O-SMGA's performance. As shown in Fig. 4, the imaging results for both low-resistivity and high-resistivity anomaly models using O-SMGA are largely similar, differing only in terms of positive and negative. Therefore, in this section's performance testing and subsequent transmitting source design, only the low-resistivity anomaly model is used.

O-SMGA theoretically has the capability to detect 3-D underground electrical anomalies. To test this ability, six additional survey lines are added to the model in Fig. 1, each running parallel to the original survey line. The x-coordinates for the new survey lines are -150, -100, -50, 50, 100, and 150 m. Magnetic field data from the original survey line and the six newly added survey lines are used to image, and the results are shown in Fig. 6.

As observed in Fig. 6, the size of the abnormal region in the imaging results tends to decrease as the distance between the survey line and the anomaly center increases. When the horizontal projection of a survey line does not intersect with the anomaly, the abnormal region disappears in the imaging results. This observation allows the determination

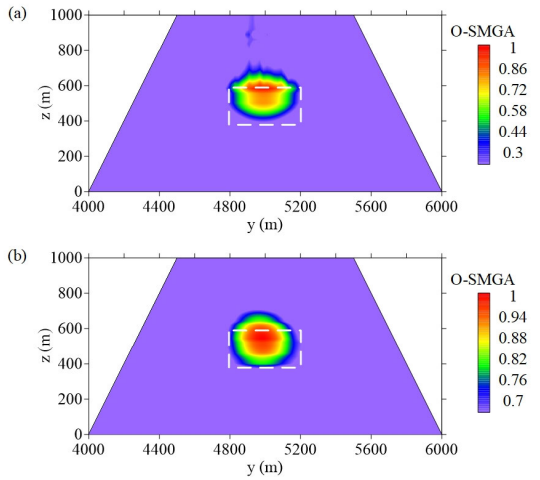


Fig. 7. Test for variations in anomaly resistivity: imaging results of O-SMGA after modifying the anomaly resistivity. (a) Anomaly resistivity is 50 Ωm. (b) Anomaly resistivity is 1 Ωm.

of the position and boundaries of the anomaly in the x, y, and z directions, enabling effective 3-D identification of underground electrical anomalies.

In the model in Fig. 1, the difference between anomaly and background resistivity is ten times. However, in actual detection, this difference may be smaller or larger than ten times. Therefore, to further validate the feasibility of the imaging method, the resistivity of the low-resistivity anomaly in the model in Fig. 1 is modified to $\rho_1 = 50$ Ωm and $\rho_1 = 1$ Ωm, respectively. The models are then reconstructed using COMSOL to obtain the magnetic field along the survey lines. Mesh and other parameters are detailed in the appendix. Imaging is performed on the models with modified anomaly resistivity, and the results are presented in Fig. 7.

Comparing Figs. 4 and 7, it is evident that when the threshold for delineating the abnormal region is kept constant (i.e., when using the same colorbar), variations in the difference between the anomaly resistivity and background resistivity do not significantly impact the position of the abnormal region in the O-SMGA imaging results. However, the boundaries of the abnormal region change: the boundaries contract when the difference decreases, and expand when the difference increases. Consequently, the boundaries of abnormal regions no longer align with those of anomaly in the model from Fig. 1. This discrepancy arises because the threshold of 0.5 used to delineate the anomaly region aligns with a resistivity difference of ten times but does not correspond to differences of 2 or 100 times. To match the boundaries in Fig. 7(a) and (b) with the anomaly boundaries in Fig. 1, the threshold would need to be decreased or increased, respectively. This observation indicates that the O-SMGA parameter reflects the extent of change in comprehensive underground resistivity caused by the presence of electrical anomalies. The greater the change, the higher the threshold required to delineate the O-SMGA abnormal region (or lower, in the case of high-resistivity anomalies). Based on this characteristic, further study can be conducted on the relationship between the calculated values of O-SMGA and the actual underground resistivity.

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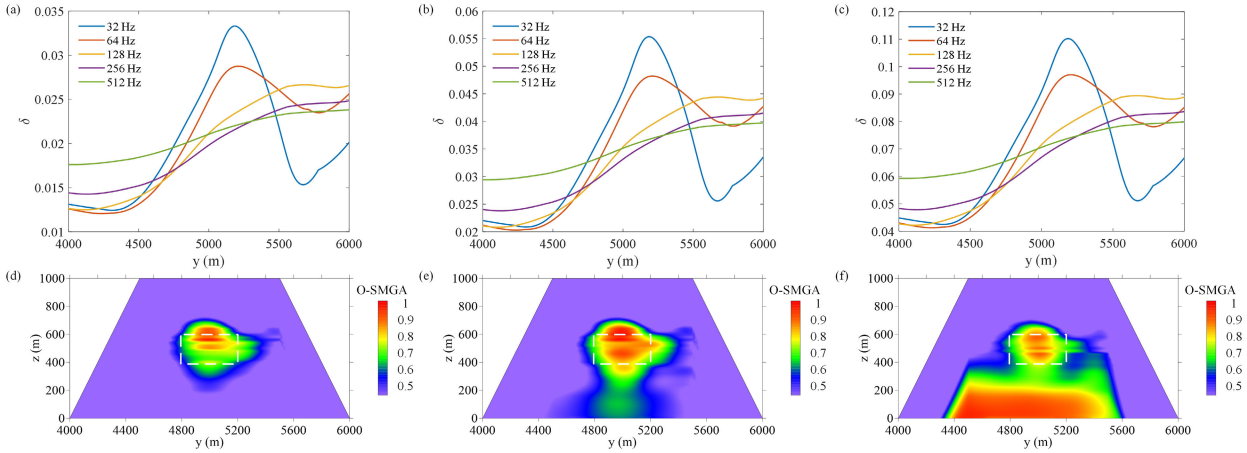


Fig. 8. Test for background resistivity deviations: influences on background field. (a) 10% deviation, (b) 15% deviation, and (c) 20% deviation. Influences on imaging results. (d) 10% deviation, (e) 15% deviation, and (f) 20% deviation.

During the imaging process with O-SMGA, the calculation of background resistivity involves ARI using a binary search iteration method to calculate the apparent resistivity at each point and then taking the mean [15]. However, the binary search iteration process may introduce deviations in the calculation of apparent resistivity due to the precision of the iteration, leading to inaccuracies in the fitting of background resistivity. In the imaging process in Fig. 4, the calculated result for background resistivity is 97.9 Ωm, deviating by 2.1% from the true background resistivity. However, in actual detection, an increase in terrain complexity may exacerbate the deviation in background resistivity calculation. Currently, the model used is a half-space model, but in actual detection, underground structures are often much more complex, which could further impact the accuracy of background resistivity calculations. Thus, it is essential to analyze the effects of background resistivity deviations on O-SMGA. A deviation in background resistivity fundamentally impacts the accuracy of the background magnetic field. Therefore, the first step is to assess how these resistivity deviations influence the background magnetic field. Using COMSOL, the background magnetic fields are obtained from the model in Fig. 1, where the background resistivity values are set at 90, 85, and 80 Ωm, respectively. The relative deviation δ between these values and the background magnetic field with a resistivity of 100 is calculated as shown in (11). Among them, $B_{02}(x_i, y_i, f_k)$ as the background magnetic field true value, $B_{03}(x_i, y_i, f_k)$ as the background magnetic field when there is a deviation of background resistivity. Plots of δ against different frequencies along the survey line are shown in Fig. 8(a)–(c)

$$\delta(x_i, y_i, f_0) = \frac{B_{03}(x_i, y_i, f_0) - B_{02}(x_i, y_i, f_0)}{B_{02}(x_i, y_i, f_0)}. \tag{11}$$

From Fig. 8(a)–(c), it is apparent that the influence of background resistivity deviation on the background magnetic field intensifies with larger deviations in background resistivity. This suggests that for O-SMGA, larger deviations in background resistivity lead to greater deviations between the imaging results and the actual underground structures. Additionally, when background resistivity deviations are equal, the background magnetic field deviation at the same position tends to decrease with increasing frequency. According to (7), it can be inferred that the O-SMGA imaging results are more impacted by background resistivity deviations at greater depths. To confirm this observation, data from the survey line in Fig. 1 are used as the observed magnetic field B, and the magnetic fields from Fig. 8(a)–(c) are used as the background magnetic field B_1. O-SMGA is then used for imaging, and the imaging results are in Fig. 8(d)–(f).

From Fig. 8(d)–(f), it can be observed that when the background resistivity deviates by 10% from the true value (90 Ωm), the imaging results show some distortion in the abnormal regions. However, the strong abnormal regions still correspond to the location of anomalies in the model. The boundaries of these strong abnormal regions align with the anomaly boundaries, effectively identifying underground electrical anomalies. When the background resistivity deviates by 15% (85 Ωm), the abnormal regions in the imaging results spread deeper. The strong abnormal region still roughly matches the settings of the anomaly in the model. With a 20% deviation in background resistivity (80 Ωm), severe false anomalies appear throughout the deeper region in the imaging result. The imaging results align with the analysis in Fig. 8(a)–(c). Overall, the O-SMGA shows a degree of resistance to background resistivity deviations. Within a certain range of deviation, O-SMGA can still effectively detect underground electrical anomalies.

In the imaging process shown in Fig. 4, the measured magnetic fields B used are all ideal data without noise. However, in actual detection, various types of noise may exist. Therefore, it is necessary to test the noise resistance of O-SMGA. White noise with amplitudes of 1/100 and 1/10 of the measured signal amplitude is added to the survey line data of the model in Fig. 1 as imaging data. O-SMGA is then used for imaging, and the imaging results is in Fig. 9.

From Fig. 9, it can be observed that when the noise amplitude is 1/100 of the measured signal amplitude, there is no significant impact on the imaging results. However, when the noise amplitude is 1/10 of the measured signal amplitude, there is noticeable distortion in the abnormal region

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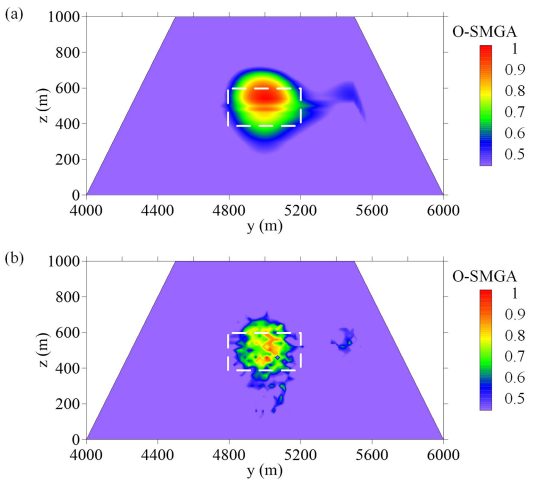


Fig. 9. Antinoise test: when the measured data contain noise, the imaging results of the O-SMGA for the model in Fig. 1. (a) When the noise amplitude is 1/100 of the measured signal amplitude. (b) When the noise amplitude is 1/10 of the measured signal amplitude.

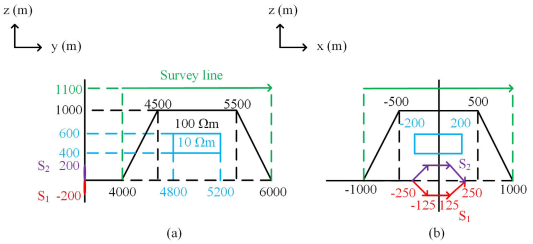


Fig. 10. When there is terrain relief in the source area, the model of GAFDEM. (a) *yz* profile and (b) *xz* profile.

of the imaging results, but it still generally conforms to the setting of anomalies in Fig. 1. For current electromagnetic detection instruments, achieving a noise amplitude not greater than 1/100 of the measured signal amplitude is relatively feasible [32]. Therefore, it can be concluded that O-SMGA exhibits good noise resistance.

In the model in Fig. 1, the transmitting source is represented as horizontal straight lines. However, in actual detection, terrain reliefs exist not only in the survey area but also in the source area. Since GAFDEM employs grounded wires as transmitting sources, terrain reliefs can cause changes in the morphology of the transmitting sources, thereby resulting in variations in the electromagnetic response of GAFDEM across the entire space. Consequently, it is necessary to analyze the impact of terrain reliefs in the source area on the imaging effectiveness of O-SMGA. Based on the model in Fig. 1, the terrain in the source area is modified, and the transmitting sources are placed flush with the ground. Using COMSOL, a model is constructed in Fig. 10 to obtain the magnetic field along the survey lines. The mesh and other parameters are detailed in the appendix. S_1 represents a concave terrain with a depth of 200 m in the source area, while S_2 represents a convex terrain with a height of 200 m. The transmitting frequency ranges from 32 to 1024 Hz with an interval of 4 Hz. O-SMGA is employed to image the models in Fig. 10, and the imaging result is shown in Fig. 11.

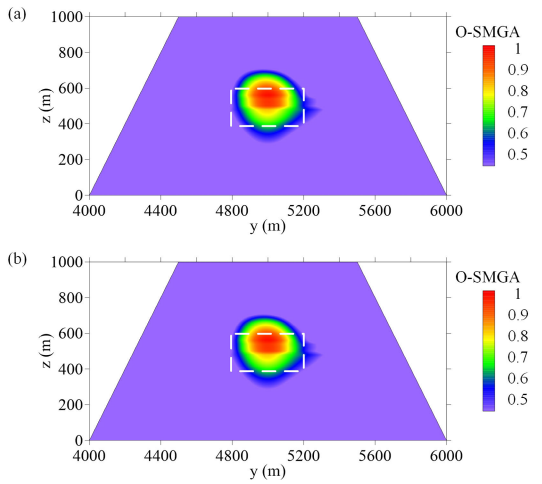


Fig. 11. Test for the terrain of the source area: The imaging results of O-SMGA for the model in Fig. 19. Transmitting source is (a) S_1 and (b) S_2.

Comparing Fig. 4 with Fig. 11, it can be concluded that the terrain relief in the source area, or the variation in the shape of the transmitting source, has little to no effect on the underground electrical anomaly identification performance of O-SMGA. This is because the COMSOL model, established with terrain information, already considers the influence of terrain when obtaining the background field correction curve, as calculated in (9). At this point, the background field correction curve contains both the terrain information from the source area and the survey area. Therefore, when the terrain information in the source area is known, the terrain relief can be considered to have no impact on the imaging effect of O-SMGA.

III. Optimization of Transmitting Source Design in Complex Terrain

GAFDEM is a controlled-source electromagnetic method. The quality of the survey fundamentally depends on the design of the observation scheme, which mainly includes the design of transmitting sources and survey lines. Survey line design is generally determined by the detection target. Transmitting sources, being placed on the ground, face challenges in surveys over complex terrains due to terrain issues. Therefore, to further improve the effectiveness of O-SMGA in actual detections, this section will analyze the effects of various parameters of transmitting sources on the imaging results of O-SMGA using the method of controlled variables under the low-resistance anomaly model in Fig. 1. Relevant impact laws will be summarized based on the analysis results, and design guidelines for GAFDEM transmitting sources applicable to O-SMGA will be proposed. Further enhancement of O-SMGA's imaging performance will be explored from the perspective of transmitting source design.

A. Transmitting Frequency

According to the basic principle of O-SMGA, for a specific detection target, it is necessary to first determine the underground depth range of the target. Then, (7) is utilized

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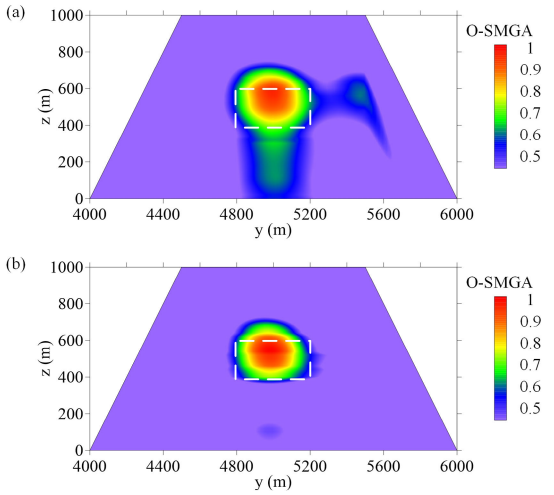


Fig. 12. Test for the transmitting frequency: the imaging results of O-SMGA for the model in Fig. 1 after changing transmitting frequency. (a) 32–1024 Hz with an interval of 4 Hz. (b) Segmented frequency method (transmitting interval: 4 Hz for 64–192 Hz, and 32 Hz for 32–64 and 192–1024 Hz).

to inversely calculate the required transmitting frequency band. Since the calculation of O-SMGA includes correction coefficients for frequency gradients, the ideal transmitting frequencies should be evenly spaced frequencies within the target frequency band. In theory, the smaller the frequency interval, the better the imaging effect of O-SMGA. However, in actual detection, the inclusion of too many frequencies in the transmitting signal due to instrument limitations can lead to a decrease in signal-to-noise ratio (SNR) for each frequency. If each frequency is to meet the SNR requirements for detection, multiple transmissions are required, which significantly impacts detection efficiency. Therefore, in order to reduce the number of transmitting frequencies without compromising the detection effect, it is necessary to analyze the effect of frequency interval on the imaging performance of O-SMGA.

For the model in Fig. 1, in order to image the entire mountain, the transmitting frequency band calculated based on (7) is determined to be 32–1024 Hz. The imaging results in Fig. 4 are obtained with a frequency interval of 4 Hz. To analyze the impact of frequency interval on O-SMGA, the imaging data from Fig. 4 are extracted at intervals of 32 Hz, maintaining a transmitting frequency band of 32–1024 Hz, but increasing the interval from 4 to 32 Hz. O-SMGA is then used to image the model in Fig. 1 with the altered transmitting frequencies. The imaging results is shown in Fig. 12(a).

A comparison between Figs. 4 and 12(a) reveals that enlarging the transmitting frequency interval causes the abnormal regions to extend deeper in the z-direction. This is attributed to (7), which results in significantly larger depth differences corresponding to the same frequency interval in deeper regions compared to shallower ones. Consequently, if anomalies are situated in deeper layers, it significantly affects the detection efficiency of O-SMGA. In actual detection, for a deep underground structure, often only a certain depth range is the main target of detection. For example, in the excavation of mountain tunnels, it is only necessary to detect the underground structure at the bottom layer depth corresponding to the tunnel. Therefore, a method of setting transmitting frequencies in segments is proposed. This method reduces the frequency interval within the frequency band corresponding to the target detection depth and increases the frequency interval outside the frequency band corresponding to the target detection depth. This method satisfies the detection requirements while reducing the number of transmitting frequencies, thereby improving actual detection efficiency.

To validate this method, based on the model in Fig. 1, the transmitting frequencies are adjusted again. The depth range where the anomalies are located, that is, $z = 400$–600 m, is determined as the target detection depth. According to (7), the frequency band corresponding to the target detection depth is determined to be 64–192 Hz. Therefore, the transmitting interval of the model in Fig. 1 is set to 4 Hz for 64–192 Hz and 32 Hz for 32–64 and 192–1024 Hz. O-SMGA is used to image the model in Fig. 1 with the transmitting frequencies set using the segmented frequency method. The imaging result is shown in Fig. 12(b). From the comparison between Fig. 12(a) and (b), it can be observed that the use of the segmented frequency method has eliminated the spreading phenomenon of the abnormal regions. The imaging results are consistent with Fig. 4, with the position and boundaries of the abnormal regions corresponding well to anomalies in the model. During the imaging process, Fig. 4 used 249 frequencies, while Fig. 12(b) used only 63 frequencies, resulting in an efficiency improvement of nearly four times. Therefore, based on the analysis results above, a transmitting frequency design guideline for O-SMGA can be proposed. The guideline suggests that, first, determine the target detection depth. Further, use (7) to determine the frequency band corresponding to the target detection depth. Finally, utilize the segmented frequency method to appropriately densify the transmitting frequencies within the frequency band corresponding to the target depth and appropriately sparse the transmitting frequencies within the frequency bands corresponding to nontarget depths, within the permissible limits of the instrument. When imaging with O-SMGA under these conditions, both the accuracy of detecting the target underground structure and the efficiency of detection can be effectively improved.

B. Transmit-Receive Distance and Transmitting Source Length

In GAFDEM's actual detection, to maximize the SNR of measured magnetic field data in a given noise environment, it is common to either decrease the transmit-receive distance or increase the electric dipole moment. The transmit-receive distance refers to the minimum distance between survey points along a survey line and the transmitting source. Due to the attenuation characteristics of electromagnetic waves produced by the transmitting source, reducing the transmit-receive distance can effectively increase the intensity of the measured magnetic field in the survey area, thereby improving SNR. Increasing the electric dipole moment can be achieved by either increasing the current in the transmitting source or lengthening the transmitting source, both of which boost the magnetic field's intensity in the survey area, thus enhancing SNR. Increasing only the current does not affect the

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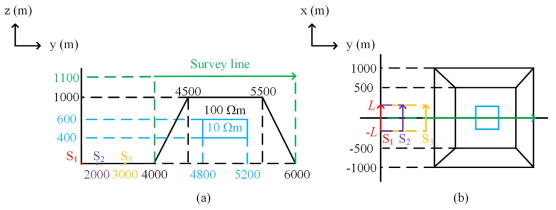


Fig. 13. Model of GAFDEM after changing the transmit-receive distance and transmitting source length. (a) *yz* profile and (b) *xy* profile.

distribution pattern of electromagnetic waves in the survey area and therefore does not impact the accuracy of O-SMGA imaging. However, for the same survey area, transmitting sources of varying lengths produce different distributions of electromagnetic waves [33]. Additionally, transmitting sources of identical lengths but with varying transmit-receive distances generate electromagnetic waves that differ in distribution [34]. Therefore, this section combines the transmit-receive distance with the length of the transmitting source to analyze their combined effects on O-SMGA imaging accuracy.

Based on the model in Fig. 1, the transmitting source is modified to have a length of $2L$ with its center coordinates at (0, 0, 0), (0, 2000, 0), and (0, 3000, 0), representing sources S_1, S_2, and S_3. Their respective transmit-receive distances are 4000, 2000, and 1000 m. L has three possible values: 500, 1000, and 2000 m. Other parameters remain unchanged. COMSOL is used to create the model as shown in Fig. 13 to acquire the magnetic field along the survey line. The grid configuration and other parameters are detailed in the appendix. The transmitting frequencies are set using the segmented frequency method (transmitting interval: 4 Hz for 64–192 Hz, and 32 Hz for 32–64 and 192–1024 Hz). O-SMGA is employed to image the models in Fig. 13, and the imaging result is shown in Fig. 14.

From Fig. 14, it can be observed that changes in transmit-receive distance and source length do not significantly affect the horizontal anomaly detection capability of O-SMGA but do greatly influence the vertical recognition of anomalies. When the transmit-receive distance is constant, increasing the source length causes the O-SMGA anomaly region to gradually extend deeper. When the source length is constant, decreasing the transmit-receive distance leads to distortion in the anomaly region, with the degree of distortion worsening as the distance decreases. This indicates that, ideally, larger transmit-receive distances and smaller source lengths yield better O-SMGA imaging results. However, with a constant transmitting current, the electromagnetic response intensity generated by the source decreases as the transmit-receive distance increases and the source length decreases [33], [34]. Practically, electromagnetic receiving systems have a minimum resolution threshold, and signal noise interference could drown out target signals if the electromagnetic response is too weak. According to Section II-D, excessive noise could also affect the imaging results of O-SMGA. Therefore, infinite increases in transmitter-receiver distances or reductions in source length are not realistic for actual detection. Based on the analysis in this section, along with the findings on noise resistance from Section II-D, the following design guidelines are proposed for optimizing the transmit-receive distance and source length for O-SMGA. The guidelines suggest first determining the minimum measurable magnetic field based on the electromagnetic receiving system's minimum resolution and noise level to meet the SNR requirements outlined in Section II-D. Then, a model of the survey area is created to calculate the thresholds for transmit-receive distance and source length that satisfy these requirements. Finally, in actual applications, the guidelines suggest maximizing the transmit-receive distance while minimizing the source length within permissible thresholds and as allowed by the terrain.

C. Transmitting Source Direction

In actual detection, the underground structures are often highly complex, rarely presenting as simple as the model in Fig. 1 where only a single electrical anomaly exists. Liu et al. [15] conducted studies on the capability of SMGA to identify multiple anomalies when operating on horizontal ground. Their findings demonstrated that SMGA is capable of simultaneously identifying multiple underground anomalies in a horizontal ground setup. However, for anomalies of the same electrical type, there are variations in the intensity of the abnormal regions in SMGA imaging. Anomalies closer to the source exhibit significantly higher intensity compared to those further away. This difference arises because when the transmitting source is vertical to the survey line, multiple electrical anomalies along the vertical transmitting direction have different horizontal distances from the source. It can result in anomalies located farther from the source being overshadowed by those closer to it, thereby remaining undetected. According to this conclusion, it is speculated that when the transmitting source is parallel to the survey line, the differences in horizontal distances between the transmitting source and various anomalies are minimized, which should mitigate this issue. To validate this conclusion and simultaneously analyze the impact of the relative orientation between the transmitting source and the survey line on O-SMGA, a mountainous multianomaly model is established by using COMSOL in Fig. 15 to obtain the magnetic field along the survey lines. Mesh and other parameters are provided in the appendix. The model in Fig. 15 has two low-resistivity electrical anomalies. The transmitting sources are placed at positions corresponding to vertical (S_1 in Fig. 15) and parallel (S_2 in Fig. 15) survey lines, respectively. The transmitting frequencies are set using the segmented frequency method (transmitting interval: 4 Hz for 64–192 Hz, and 32 Hz for 32–64 and 192–1024 Hz). The survey line data are extracted separately for the transmitting sources located at S_1 and S_2 to calculate the O-SMGA. The O-SMGA comparison for different source positions is in Fig. 16. In the figure, O-SMGA represents the calculated values with a transmitting frequency of 100 Hz. The green dashed lines in the figure denote the positions of the anomalies.

From Fig. 16, it can be observed that when the transmitting source is parallel to the survey line, the maximum O-SMGA at the positions of the two anomalies are nearly identical, with a difference of only 0.1%. However, when the transmitting source is vertical to the survey line, there

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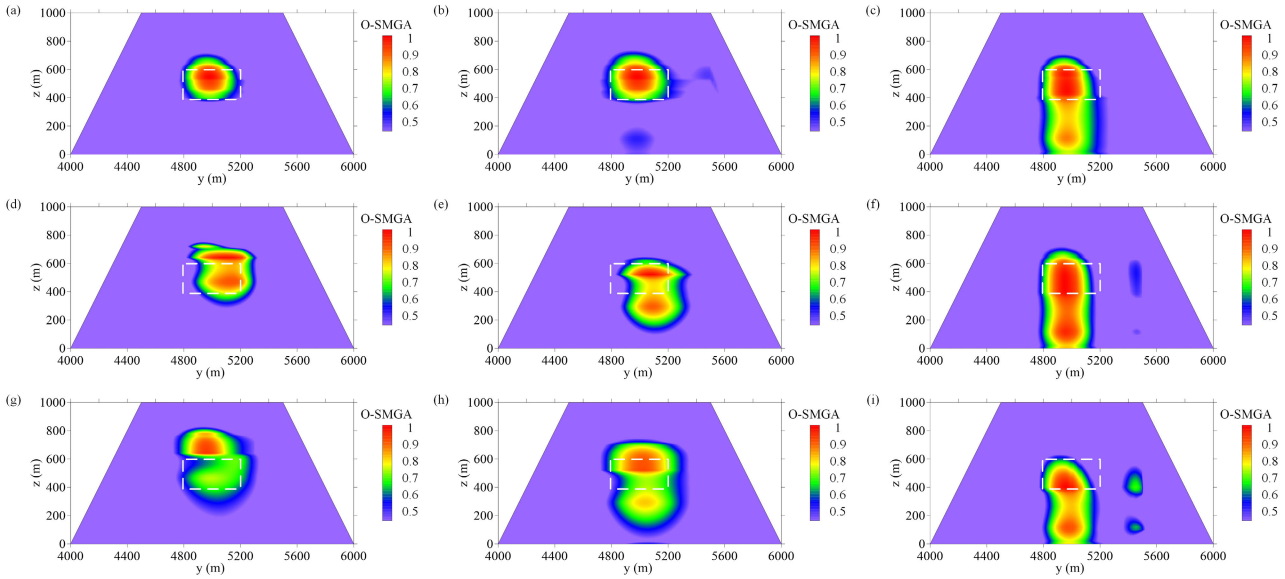


Fig. 14. Testing transmit-receive distance and source length: the transmitting source is S_1. (a) $L = 500$ m, (b) $L = 1000$ m, and (c) $L = 2000$ m. The transmitting source is S_2. (d) $L = 500$ m, (e) $L = 1000$ m, and (f) $L = 2000$ m. The transmitting source is S_3. (g) $L = 500$ m, (h) $L = 1000$ m, and (i) $L = 2000$ m.

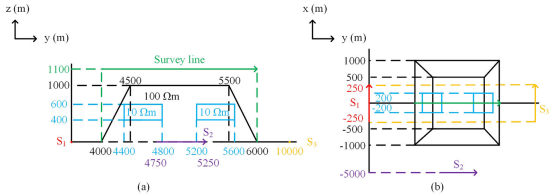


Fig. 15. Model of GAFDEM in mountainous terrain with multiple anomalies. (a) *yz* profile and (b) *xy* profile.

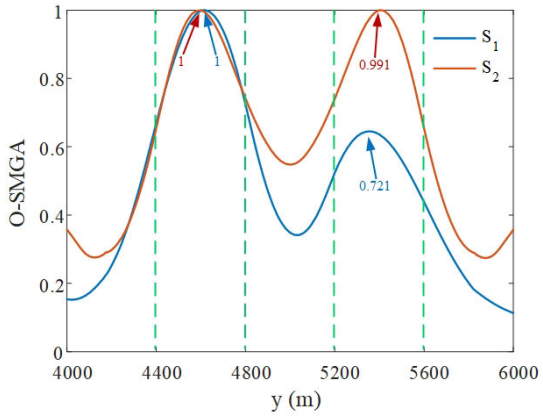


Fig. 16. When the transmitting source is positioned at different locations, the O-SMGA in the same survey line.

is a significant difference in the maximum O-SMGA at the positions of the two anomalies, with a difference of 35.1%. This result confirms that the transmitting source parallel to the survey line can effectively resolve the inconsistency in multiple anomaly identification. However, in actual detection, it may not always be feasible to deploy transmitting sources parallel to the survey line due to terrain constraints. Therefore, to address the deviation issue caused by multiple anomalies when the transmitting source is vertical to the survey line,

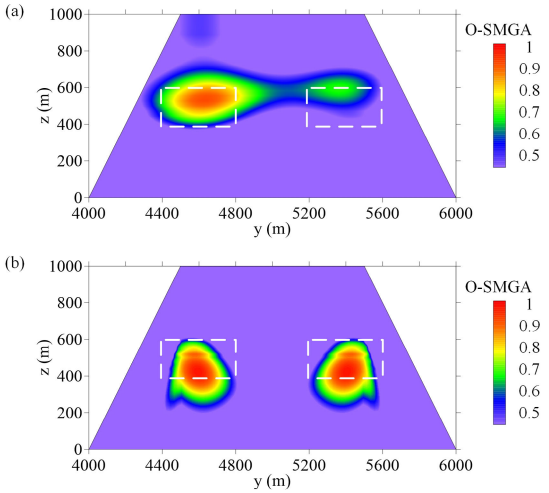


Fig. 17. Multianomaly detection capability test: the imaging results of O-SMGA for the model in Fig. 15. (a) Only S_1 exists. (b) S_1 and S_3 exist simultaneously.

a multisource combination transmitting source design method is proposed. This method suggests placing two or more transmitting sources symmetrically on both sides of the survey line, with the midpoint of the survey line as the reference point. This arrangement aims to counteract the deviation caused by the horizontal distance between multiple anomalies and the transmitting source.

To validate the effectiveness of the multisource combination transmitting source design method, an additional transmitting source, denoted as S_3, is added symmetrically to S_1, with the midpoint of the survey line (0, 5000, 1100) as the reference point. To compare the differences between the multisource combination and single-source, O-SMGA is employed to image both the model with only S_1 and the model with both S_1 and S_3. The imaging results are illustrated in Fig. 17.

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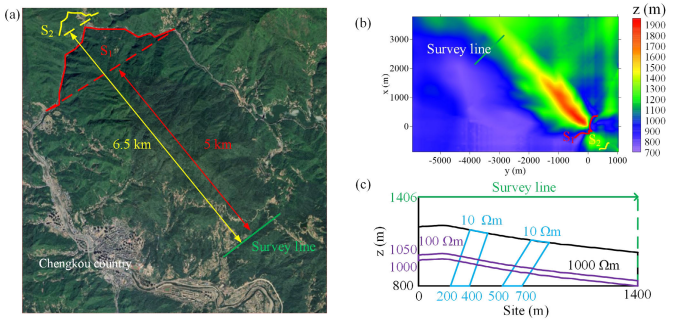


Fig. 18. After loading the actual terrain, the model of GAFDEM. (a) 3-D terrain of the survey area, (b) terrain distribution of the survey area, and (c) vertical profile of survey line.

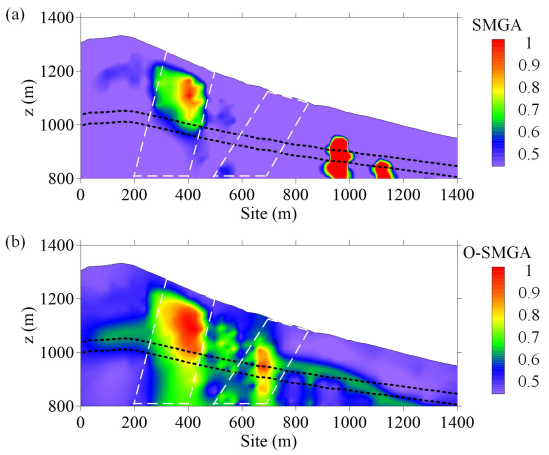


Fig. 19. Effectiveness test of O-SMGA: comparison of the imaging results for the model in Fig. 18. (a) SMGA and (b) O-SMGA.

From Fig. 17(a), it can be observed that when only one transmitting source is present, O-SMGA still exhibits inconsistent recognition of two identical electrical anomalies. However, when employing the multisource combination method, this issue is resolved. The intensities of abnormal regions with identical electrical types are nearly identical. The positions and boundaries of the abnormal regions closely match those of the anomalies in the model. The multisource combination design effectively eliminates discrepancies in O-SMGA multianomaly identification when the transmitting source is perpendicular to the survey line. However, in actual applications, it may be challenging to implement such multisource setups due to terrain constraints. Additionally, multiple transmitting sources can lead to issues such as mutual interference or phase inconsistency [17]. Based on this section's analysis, we propose a design guideline for the orientation of the transmitting source. The guideline suggests aligning the source parallel to the survey line wherever possible. If only a perpendicular setup is feasible, this section's multisource combination method should be used to mitigate the inconsistency in multianomaly identification.

IV. Synthetic Test

To further validate the actual effectiveness of the theoretical research presented in this article, this section analyzes the performance of O-SMGA and the transmitting source design guidelines using data from a model that loads actual terrain data, along with field data from actual experiments.

A. Effectiveness Analysis of O-SMGA

Terrain data near Chengkou, Chongqing, China, is obtained using BIGEMAP and loaded into COMSOL to establish an equivalent model, as shown in Fig. 18(a) [35]. The mesh and other parameters are provided in the appendix. Additionally, based on actual detection conditions, transmitting sources and survey lines are set up. The terrain distribution, transmitting source settings, survey line settings, and underground anomaly settings are shown in Fig. 18(b) and (c). Fig. 18(b) illustrates the height distribution of the region containing the transmitting sources S_1 and survey lines. In the figure, the red lines represent the transmitting sources. To better simulate actual detection, transmitting frequencies of 32, 64, 128, 256, 512, 1024, 2048, 4096, and 8192 Hz are used. Fig. 18(c) shows the vertical profile of the survey lines, with the horizontal axis representing the positions of various survey points along the profile. The black lines represent the terrain relief of the survey area, while the blue lines represent anomalies. In actual detection, according to geological data and the results of other geophysical methods, the resistivity of the earth in this area is approximately 1000 Ωm. Additionally, there is a water-bearing fault and a water-bearing fracture, both of which are low-resistance anomalies. Therefore, in Fig. 18, the background resistivity, that is, the resistivity of the earth, is set to 1000 Ωm, and the resistivity of the two low-resistance anomalies is set to 10 Ωm. Furthermore, to simulate actual complex geological structures, an abnormal layer with a resistivity of 100 Ωm is added to the mountain, as shown by the purple line in Fig. 18(c). The green line in Fig. 18 serves as the survey line. The magnetic field amplitude of each survey point on the survey line is calculated using COMSOL as the acquisition data for GAFDEM. SMGA and O-SMGA are used to image the model in Fig. 18, and the imaging results are shown in Fig. 19.

Comparing Fig. 19(a) and (b), it is evident that O-SMGA exhibits significantly better capability to identify underground electrical anomalies in complex terrain compared to SMGA. In the SMGA imaging results, the anomaly between the 500 and 700 m positions remains undetected, and only the anomaly between the 200 and 400 m positions can be identified. Furthermore, in Fig. 19(a), the presence of the abnormal layer in Fig. 18 is not discernible. However, in the O-SMGA imaging results, both anomalies are effectively identified, with their positions aligning closely with the anomaly centers in Fig. 18. The spread of the anomalies aligns with the expected direction of their trend. The anomalies appear as positive regions in the O-SMGA results, indicating low-resistivity anomaly, which aligns with the model. Additionally, the presence of the abnormal layer is detectable in the O-SMGA imaging results, providing a rough estimate of the layer's depth and orientation. Overall, these results suggest that compared to SMGA, O-SMGA offers an improved ability to detect underground electrical anomalies in complex terrain.

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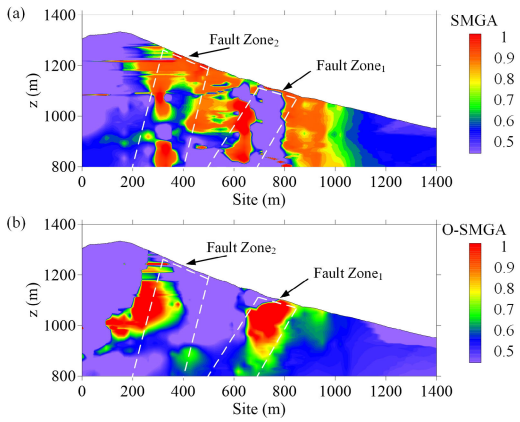


Fig. 20. Effectiveness test of O-SMGA: comparison of imaging results in the field experiment. (a) SMGA and (b) O-SMGA.

To further validate the effectiveness of O-SMGA in the actual field, a field experiment was conducted. The test area is located near Chengkou County in Chongqing, China, the same location used for the model in Fig. 18. The parameters for the transmitting source, survey lines, and transmitting frequencies are consistent with those in Fig. 18. First, SMGA is applied for imaging. During the imaging process, the background field used was the same as that in Fig. 19(a). The imaging result is in Fig. 20(a). The figure illustrates scattered positive abnormal regions. According to the definition of SMGA, these abnormal regions would typically be interpreted as low-resistivity anomalies. However, based on actual geological data and results from other detection methods [35], it is known that the locations marked by white lines in the figure correspond to two water-bearing faults, both of which are low-resistivity anomalies. Specifically, fault zone$_1$ corresponds to the Chengba Fault Zone, while fault zone$_2$ corresponds to an unnamed fault. Significant differences exist between the positions and boundaries of the abnormal regions in the imaging results of Fig. 20(a) and those of the two fault zones.

Furthermore, O-SMGA imaging is conducted. The imaging result is presented in Fig. 20(b). During the imaging process, the background field utilized is derived from the imaging process of Fig. 19(b). From Fig. 20(b), it can be observed that there are two positive abnormal regions. At the location of fault zone$_1$, the position and boundary of the abnormal region closely match those of the fault. At the location of fault zone$_2$, the position of the abnormal region is shifted to the left, and the boundary in the shallow portion resembles that of the fault. The imaging results in Fig. 20(b) are markedly superior to those in Fig. 20(a), and they align well with the known data, providing a macroscopic reflection of the positions and orientations of the fault and the fault zone. Therefore, the effectiveness of O-SMGA can be validated from Fig. 20. This demonstrates that O-SMGA possesses the capability to identify underground electrical anomalies in actual complex terrains.

B. Effectiveness Analysis of Transmitting Source Design Guidelines

To validate the effectiveness of the transmitting source design guidelines in Section III, the transmitting source of the model in Fig. 18 is optimized based on these guidelines. The original transmitting source (S_1) is replaced with S_2, depicted by the yellow line in Fig. 18. Compared to S_1, S_2 has a shorter length and a greater transmit-receive distance. The transmitting frequency remained unchanged. Imaging is conducted using O-SMGA, with the results shown in Fig. 21.

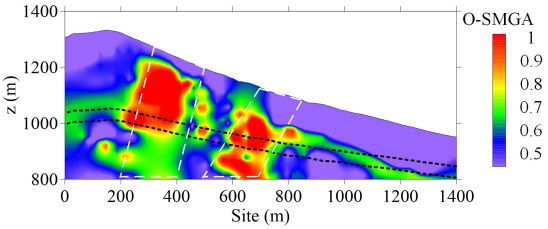


Fig. 21. Effectiveness test of transmitting source design guideline: O-SMGA imaging results for the model in Fig. 18 after transmitting source optimization.

A comparison between Figs. 19(b) and 21 reveals that after optimizing the transmitting source design, the boundaries of the high-intensity abnormal region in the imaging results align more closely with the horizontal boundaries of the anomaly in the model, and the direction of deep spreading of the high-intensity region also matches the anomaly. Additionally, compared to Fig. 19(b), the strip-like abnormal region in Fig. 21, caused by the abnormal layer, exhibits a higher intensity, aligning better with the layer's location and direction. This comparison indicates that the proposed transmitting source design guidelines can be effectively applied in actual complex terrain areas to improve the performance of O-SMGA.

V. Discussion

This article analyzed the applicability of O-SMGA in multiple models and proposed optimization guidelines for O-SMGA transmitting sources. First, the feasibility of O-SMGA in terrain reliefs is verified (Fig. 4). Furthermore, the following conclusions are drawn from the performance testing of the algorithm.

1) O-SMGA demonstrates an effective capability for 3-D detection of underground electrical anomalies.

2) The threshold used to delineate the abnormal region in O-SMGA's imaging results has a positive correlation with the difference between background resistivity and anomaly resistivity.

3) The impact of background resistivity deviation on O-SMGA tends to increase with depth and with larger deviation magnitudes. Within a certain threshold, this impact can be considered negligible.

4) O-SMGA shows good resistance to noise.

5) When the terrain of the source area is known, the source area's terrain does not affect the imaging performance of O-SMGA.

Then, the impact of transmitting parameters on the imaging performance of O-SMGA is analyzed, leading to the following guidelines for GAFDEM transmitting source design suitable for O-SMGA: 1) the selection of transmitting frequencies should adopt a segmented frequency method, increased density in critical frequency bands when instrumentation permits;

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2) for the choice of transmit-receive distance and transmitting length, maximize the distance and minimize the length, provided that the SNR requirements are met; and 3) when selecting the transmitting source direction, aim to have the transmitting source parallel to the survey line, if terrain conditions allow. If parallel placement is not feasible due to terrain constraints, a multisource combination method can be used, placing two transmitting sources perpendicular to the survey line, symmetrically about the survey line center. Finally, using the actual field data and the model that loads actual terrain, the effectiveness of O-SMGA and the proposed transmitting source design guidelines is comprehensively tested. The results indicate: 1) O-SMGA can effectively identify the location, boundary, and electrical type of underground anomalies in actual complex terrain and 2) following the transmitting source design guidelines can further enhance the accuracy of anomaly detection.

In the performance testing of O-SMGA, the actual terrain in Fig. 18 matches the surveyed area in the field experiment. However, there are some differences between the imaging results in Figs. 19(b) and 20(b), such as the offset in the location of abnormal regions at 200–400 m in Fig. 19(b). Taking into account the instrument performance used in the field experiment, several factors contribute to this situation.

1) According to (1), the space gradient *Grad* of the actual magnetic field is calculated based on the differences in magnetic field between survey points. This calculation method makes *Grad* less susceptible to short-term, nondrastic noise. However, in actual detection, since the electromagnetic receiving system is mounted on a flying platform, it may introduce significant motion and attitude noise into the measured signal. This noise, influenced by factors such as weather and flight altitude, exhibits characteristics of short-term drastic changes, which can affect the calculation of *Grad* and thus impact the imaging effect of O-SMGA.

2) According to (1), the calculation of *Grad* involves the spacing between survey points. In actual detection, this spacing is determined by calculating the latitude and longitude of each survey point using GPS. The GPS accuracy used in the experiment ranges from 3 to 7 m, leading to deviations from the true positions. This deviation can affect the calculation of *Grad* and subsequently influence the imaging effect of O-SMGA. Moreover, this deviation can also affect the calculation of (7), resulting in errors in the coordinate transformation for imaging. The offset in the location of abnormal regions at the 200–400 m in Fig. 20(b) may be attributed to this issue. Theoretically, the model represented in Fig. 18 and the surveyed area in the field experiment are the same. The same background field is utilized for imaging in Figs. 19(b) and 20(b). However, when constructing the model in Fig. 18, the terrain information for the entire surveyed area is obtained by interpolating the terrain data extracted at fixed intervals from BIGEMAP. As a result, there are inherent differences between the terrain model in Fig. 18 and the actual terrain, leading to differences between the calculated background field and the actual background field, thereby impacting the imaging effectiveness.

Based on the study conducted in this article and the results of field experiments, several limitations of O-SMGA in actual detection can be summarized.

1) Currently, the imaging results of O-SMGA can only determine the location, boundary, and type of underground electrical anomalies, without being able to directly recover the true electrical resistivity underground like ARI. However, as indicated in Section II-D of this article, there exists a relationship between the difference in anomaly resistivity and background resistivity, and the intensity of abnormal region. With this characteristic in mind, it is hopeful to transform O-SMGA into a method for estimating electrical resistivity, which is one of the future study directions of this study.

2) Short-term, drastic fluctuations in noise can affect imaging results. It is advisable to consider suppressing such noise from the perspectives of instrument design or data processing.

3) Deviations in the positions of survey points can affect imaging results. Therefore, it is recommended to use high-precision position survey devices whenever possible.

4) The accuracy of the modeling of the surveyed area can affect imaging results significantly. This issue is the most crucial factor affecting the imaging effectiveness of O-SMGA. Hence, when modeling the surveyed area, it is important to strive for higher precision while balancing the requirements of calculation performance and calculation time.

VI. Conclusion

To meet the demand for detecting deep underground structures in complex terrain, this study initially analyzed the feasibility of SMGA in areas characterized by significant terrain variations using a simplified mountain model. Based on the analysis results, it is identified that SMGA encounters challenges with background field calculation errors in complex terrain, prompting the proposal of a corresponding method for rapid background field calculation. This method corrected the errors introduced by the terrain in background field calculation while maintaining a short time, aligning with the requirements of rapid field detection. Subsequently, performance testing is conducted on O-SMGA. Within a certain threshold, the method is less impacted by actual factors such as background resistivity deviation and noise. It demonstrates effective 3-D underground anomaly detection capabilities and holds the potential for extracting underground physical parameters. Furthermore, an analysis is conducted on the effects of various parameters of the transmitting source on the imaging performance of O-SMGA. Based on the analysis results, a design guideline of the transmitting source applicable to O-SMGA is proposed to further enhance its application capabilities in actual detection.

Finally, the synthetic tests are conducted on the O-SMGA method and the transmitting source design guidelines. The results indicate that O-SMGA retains strong anomaly detection capabilities in actual complex terrains, with accuracy that can be further enhanced by applying the transmitting source design guidelines. In comparison to traditional GAFDEM, this study fully accounted for the influence of terrain on imaging methods and the limitations of transmitting source design methods.

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TABLE I
COMSOL SETTING PARAMETERS OF ALL MODELS IN THIS ARTICLE

Name	Parameters
Version	COMSOL Multiphysics5.6
Module	AC/DC
Physics	Magnetic Field (mf)
Study	Frequency Domain
Solver	BiCGStab

TABLE II
MESH AND MATERIAL PARAMETERS OF ALL MODELS IN THIS ARTICLE

Medium	Mesh (m)	Resistivity (Ωm)
Air	$\leqslant 200$	1×10^{7}
Earth	$\leqslant 100$	100 (except for Fig.3 and Fig.18) 100, 1000, 2000 (in Fig.3) 1000 (in Fig.18)
Anomaly	$\leqslant 20$	Low-resistance anomaly: 10 (except for Fig.7) Low-resistance anomaly: 50, 1 (in Fig.77) High-resistance anomaly: 1000
Abnormal layer (only in Fig.18)	$\leqslant 50$	100
Transmitting source	$\leqslant 20$	N/A
Survey line	$\leqslant 20$	N/A

It introduces new technologies for GAFDEM in complex terrain, thereby further improving its application effectiveness in actual detection.

APPENDIX

This section provides related settings and the parameters used for modeling in COMSOL. Table I shows the COMSOL version used, the module used, and other parameters. The geometric parameters of the models are illustrated in Figs. 1, 10, 13, 15, and 18. Table II displays the mesh and material parameters for all models. All models employ similar meshes for similar media (e.g., Earth, air), as in Table II. The relative permeability and relative permittivity of each medium in all of the models are set to 1 in the material parameters. The resistivities are listed in Table I.

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